Digital Object Identifier

Agentic Quantum-Safe Privacy Framework (AQSPF): A Multimodal AI Approach to Quantum-Resilient Data Protection

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The implementation evidence and source code for this paper are available at
https://drive.google.com/drive/folders/1TqMvqNkHZtyQxz7UKHxHSbr6f1Lqs0Dj?usp=sharing.

ABSTRACT Social media exposes users to privacy risks through inadvertent sharing of personally identifiable information (PII). Existing moderation tools emphasize offensive-content detection but rarely address privacy leaks, especially when identifiers are hidden in multimodal posts. This paper presents an agentic AI framework that autonomously detects and anonymizes PII using deep transfer learning, multimodal named-entity recognition (MNER), and quantum-safe cryptography. The proposed Agentic Quantum-Safe Privacy Framework (AQSPF) integrates quantum transfer learning (QTL) for low-confidence cases and lattice-based post-quantum cryptography (PQC) for secure anonymized storage. Experiments show that combining privacy preservation, multimodal reasoning, and quantum resilience yields a practical path toward ethical social-media ecosystems.

INDEX TERMS Privacy preservation, multimodal learning, named-entity recognition, quantum transfer learning, post-quantum cryptography, agentic AI

I. INTRODUCTION

In the modern digital era, social media platforms have become a ubiquitous medium for communication, personal expression, and data sharing. Billions of users around the world continuously upload posts, photos, videos, and other multimedia content to platforms such as X (formerly Twitter), Instagram, Reddit, and Facebook. However, the exponential growth in social data sharing has created a parallel rise in privacy vulnerabilities, especially the inadvertent exposure of Personally Identifiable Information (PII). PII includes any data that can uniquely identify an individual—such as names, email addresses, phone numbers, bank details, biometric identifiers, and location metadata. Even seemingly harmless posts, when combined with contextual clues, can lead to identity inference and data breaches.

Conventional content moderation systems primarily focus on detecting toxicity, hate speech, or misinformation. Their pipelines are optimized for semantic toxicity classification or sentiment analysis but not for privacy preservation. These models fail to recognize implicit or obfuscated identifiers—for example, users writing "ph0ne num: 9876xxxxxx" or embedding screenshots of contact details in images. Moreover, social media data is increasingly multimodal, combining text, images, audio, and metadata, which complicates the task of privacy-aware AI systems. Traditional NLP-based detectors, built only on textual data, cannot interpret entities embedded within visual modalities such as profile pictures, memes, or scanned documents.

The combination of high data volume, heterogeneous modalities, and adversarial obfuscation techniques calls for a new paradigm in privacy management—one that moves beyond reactive filtering to proactive, agentic, and quantum-secure automation. This study introduces an **Agentic AI Framework for Privacy-Preserving Information Extraction Using Multimodal and Quantum-Safe Architectures**, referred to as the **Agentic Quantum-Safe Privacy Framework (AQSPF)**. The proposed system aims to autonomously detect, anonymize, and securely store PII within social media content streams by integrating Deep Transfer Learning (DTL), Multimodal Named Entity Recognition (MNER),

Quantum Transfer Learning (QTL), and Post-Quantum Cryptography (PQC) in a unified, self-regulating architecture.

The motivation for this work arises from three fundamental gaps in current research and practice:

1) **Absence of holistic privacy automation** – Current privacy-preserving mechanisms act as passive filters or rule-based scanners. They lack adaptability to new data formats and cannot autonomously learn user-specific privacy preferences.
2) **Inadequacy of unimodal analysis** – Most PII detection approaches analyze text in isolation, disregarding the visual context or multimodal correlations that frequently encode hidden identifiers.
3) **Emerging quantum threats to data security** – With the progress in quantum computing, conventional encryption standards such as RSA and ECC are at risk of decryption by quantum algorithms like Shor's and Grover's. Hence, data privacy systems must evolve towards quantum-resilient protection.

To address these issues, the proposed AQSPF employs an agentic architecture, where the AI system functions as an autonomous privacy agent capable of learning, decision-making, and self-adaptation. This agent continuously monitors multimodal data streams, detects sensitive content, determines appropriate anonymization strategies, and applies post-quantum cryptographic methods for secure archival. The system maintains human-in-the-loop supervision, ensuring transparency and explainability in every privacy decision.

A high-level system overview is illustrated in Fig. 1, depicting the flow from input data acquisition to secure storage:

- **Textual Layer:** Extracts and classifies textual PII entities such as names, addresses, and credentials using fine-tuned BERT/DeBERTa models.
- **Multimodal Layer:** Performs entity alignment between text and image modalities using a CLIP-based Graph Neural Network (GNN) fusion mechanism.
- **Quantum Layer:** Engages a Variational Quantum Circuit (VQC) for uncertainty-driven refinement when detection confidence is below a defined threshold.
- **PQC Layer:** Encrypts the anonymized outputs using lattice-based Kyber (for key exchange) and Dilithium (for digital signatures), ensuring quantum-safe persistence.

This workflow illustrates a seamless cross-layer orchestration between perception (AI detection) and protection (quantum-safe encryption). The agentic nature allows for context-aware decision-making, while the quantum-safe component ensures long-term data security, even in the post-quantum era.

From a theoretical standpoint, privacy preservation introduces a measurable trade-off between privacy protection and model performance. Overly strict anonymization can degrade model accuracy by removing contextual clues, while insufficient privacy preservation exposes sensitive data. This

FIGURE 1. Textual PII Detection Result using BERT NER.

balance is mathematically modeled by the privacy–accuracy trade-off equation, defined as:

$$P = \frac{1}{1 + e^{-\alpha(A - A_0)}} \tag{1}$$

where P represents the privacy level achieved, A is the model's achieved accuracy, A_0 denotes the target baseline accuracy, and α is a sensitivity coefficient controlling the trade-off curve. A higher α shifts the equilibrium toward privacy prioritization, while a lower value favors accuracy. This relationship forms the foundation of adaptive thresholding in AQSPF, dynamically adjusting anonymization aggressiveness based on model confidence and user-defined privacy policies.

The novelty of AQSPF lies in combining AI agent autonomy, multimodal context understanding, and quantum-era resilience into one scalable framework. The agentic approach allows proactive decision-making—analyzing intent, content, and modality coherence to infer potential privacy risks before disclosure occurs. The multimodal architecture enhances robustness against obfuscated identifiers, while the quantum layer ensures that sensitive features are securely transformed in quantum latent space before encryption.

Furthermore, the framework is designed for interoperability and deployment scalability. The modular design allows integration into existing social media APIs or enterprise data governance systems. Each module (Text, Multimodal, Quantum, PQC) can operate independently or synergistically under the agent's policy engine. The framework leverages containerized microservices and federated learning protocols to maintain privacy compliance without centralized raw data aggregation.

This study's research objectives can be summarized as follows:

1) To develop a multimodal, agentic AI model capable of extracting PII across text and image modalities.
2) To incorporate quantum transfer learning for uncertainty-based feature refinement.

3) To establish post-quantum cryptography integration for secure anonymization and storage of extracted PII.
4) To evaluate the privacy–accuracy balance using controlled experiments on multimodal social datasets.

Through these contributions, the proposed AQSPF demonstrates a future-ready architecture capable of defending digital privacy not only through advanced detection intelligence but also by embedding quantum-safe encryption at its core. This integration of cognitive AI reasoning and quantum cryptographic security positions the framework as a pioneering step toward ethical, privacy-preserving automation in the era of large-scale multimodal data.

II. BACKGROUND AND RELATED WORK

The foundation of privacy-preserving artificial intelligence lies at the intersection of natural-language understanding, multimodal perception, and cryptographic resilience. In recent years, researchers have moved from static rule-based detection toward self-adapting transformer frameworks and quantum-safe computation models. This section surveys the state of the art across four research pillars relevant to the proposed **Agentic Quantum-Safe Privacy Framework (AQSPF)**: (A) PII extraction, (B) multimodal NER, (C) quantum-enabled learning, and (D) post-quantum cryptography.

A. PII EXTRACTION USING DEEP TRANSFORMERS

Early privacy-detection systems relied on handcrafted regular expressions and support-vector classifiers. Such approaches, though precise for explicit tokens like phone numbers, failed to identify contextual or obfuscated PII.

The advent of transformer-based encoders revolutionized this task. Models such as BERT, RoBERTa, and DeBERTa generate contextual embeddings that capture syntactic and semantic dependencies across entire sentences. Liu *et al.* (2021) introduced a Deep Transfer Learning PII Extraction (DTL-PIIE) model that fine-tuned BERT on anonymized Twitter datasets, achieving approximately 97% F1-score. Shahriar *et al.* (2024) extended this work with comparative transformer analysis, revealing that RoBERTa's dynamic masking improved recall on noisy PII tokens by about 3%. Kulkarni and Cauvery (2021) combined unsupervised K-means clustering with BERT embeddings, enabling latent PII discovery without labeled data.

Despite these successes, these text-only systems ignore multimodal cues such as screenshots or embedded EXIF data, leaving significant privacy gaps. The fundamental mathematical mechanism driving such contextual learning is the *scaled dot-product attention*, formulated as

$$A = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

where Q, K, and V denote the query, key, and value matrices, respectively, and d_k is the dimensionality of the key vectors. Equation (2) enables transformers to assign differential importance to tokens, forming the contextual backbone for PII recognition in AQSPF's textual module.

B. MULTIMODAL NAMED ENTITY RECOGNITION (MNER)

Modern social platforms blend textual and visual content, demanding multimodal fusion. Named-Entity Recognition (NER) extended to multimodal inputs—MNER—detects entities appearing across captions, hashtags, and images.

Zheng *et al.* (2021) proposed an Object-Aware MNER model using adversarial gated bilinear attention to fuse visual object features with text embeddings. Liu *et al.* (2021) introduced Uncertainty-Aware MNER (UAMNer), where Bayesian estimation mitigates noise from irrelevant images. Ding *et al.* (2024) advanced this with Entity-Label-Guided Graph Fusion, modeling semantic relations between visual and textual nodes.

These models collectively achieve superior F1 scores on benchmark datasets such as Twitter2015/2017 and SNAP but lack downstream privacy control. They extract entities without performing any protective action (such as redaction or encryption). AQSPF bridges this gap by embedding MNER inside an agentic loop that automatically anonymizes and encrypts outputs.

C. AGENTIC PRIVACY MANAGEMENT

The term *Agentic AI* refers to autonomous systems capable of goal-oriented decision-making and policy learning. In the privacy domain, agentic approaches enable context-aware data handling through continuous feedback.

Kurtan and Yolum (2020) developed a privacy assistant that learns user sharing patterns and recommends safer settings. Ben Salem *et al.* (2022) introduced privacy-preserving nudges, combining automation with human oversight. Kaneko *et al.* (2023) implemented Docker-vectorized agents for scalable cloud privacy management.

However, these systems remain reactive or advisory. In contrast, the proposed AQSPF is proactive and self-regulating—detecting, anonymizing, and encrypting without manual triggers, while maintaining a human-in-loop audit layer for explainability. This transition from human-assisted to autonomous privacy constitutes a significant step toward practical, ethical AI.

D. QUANTUM MACHINE LEARNING (QML) FOR PRIVACY AND NLP

Classical deep learning models operate within deterministic vector spaces, but quantum machine learning (QML) introduces superposition-based representations that encode probabilistic states. Hybrid classical–quantum architectures can capture richer decision boundaries in low-confidence regions of data.

Ardeshir-Larijani and Fatmehsari (2024) demonstrated Hybrid Quantum Transfer Learning (QTL) for text classification, linking BERT embeddings with Variational Quantum

Circuits (VQC). Buonaiuto *et al.* (2024) applied similar hybrids to linguistic acceptability tasks, improving robustness against adversarial noise. Chen *et al.* (2022) developed Quantum LSTM (QLSTM) to model temporal dependencies in language sequences. These approaches motivate AQSPF's quantum module, which enhances borderline PII predictions and supports secure feature encoding before encryption.

E. POST-QUANTUM CRYPTOGRAPHY (PQC)

While AI techniques strengthen detection, long-term data security relies on cryptographic resilience. Conventional RSA and ECC protocols are vulnerable to Shor's and Grover's quantum algorithms. NIST's standardization of lattice-based schemes such as Kyber (for key encapsulation) and Dilithium (for digital signatures) marks the transition toward quantum-safe communication.

Dekkaki *et al.* (2024) surveyed the migration path to PQC and highlighted Kyber's efficiency in cloud architectures. Rane (2025) analyzed lattice-, code-, and hash-based algorithms, confirming Kyber and Dilithium as optimal for low-latency systems. Anastasova *et al.* (2022) proposed PQ-HPKE, a hybrid public-key scheme combining Kyber with elliptic-curve cryptography. Ranganathan and Karthikeyan (2024) layered Quantum Key Distribution (QKD) with PQC for dual-channel security.

Within AQSPF, the PQC module encrypts redacted outputs post-processing, ensuring that even if the data stream is intercepted, quantum adversaries cannot recover original PII. This integration closes the loop between AI-based detection and cryptographic protection.

F. SYNTHESIS

The review reveals a progression from syntactic detection to context-aware interpretation and ultimately to autonomous privacy decision-making with quantum-level security. Each discipline—transformer-based NLP, multimodal fusion, agentic governance, and PQC—has achieved isolated success, yet few systems unify them into a single end-to-end architecture.

The proposed AQSPF embodies that integration, serving as a bridge between semantic understanding and cryptographic assurance. Its contribution is twofold: (i) advancing the technical frontier of privacy-aware AI by fusing multimodal transformers with quantum learning; and (ii) ensuring future-proof data security through lattice-based encryption. These advances set the stage for the next section, which details the system architecture and inter-module workflow of AQSPF.

III. SYSTEM ARCHITECTURE

The proposed **Agentic Quantum-Safe Privacy Framework (AQSPF)** is an end-to-end architecture designed to autonomously detect, anonymize, and securely store Personally Identifiable Information (PII) from multimodal social media data. It unites four functional pillars—AI-driven perception, agentic decision-making, quantum-enhanced learning, and post-quantum cryptography—within a modular, containerized ecosystem. Each module is orchestrated by an intelligent privacy agent that continuously learns, adapts, and enforces security decisions.

Fig. 2 illustrates the layered workflow of AQSPF, consisting of eight interconnected modules: (1) Data Acquisition, (2) Preprocessing and OCR, (3) Text PII Detection, (4) Multimodal Fusion, (5) Quantum Enhancement, (6) Agentic Policy Engine, (7) Post-Quantum Cryptographic (PQC) Layer, and (8) Secure Storage and Audit Layer.

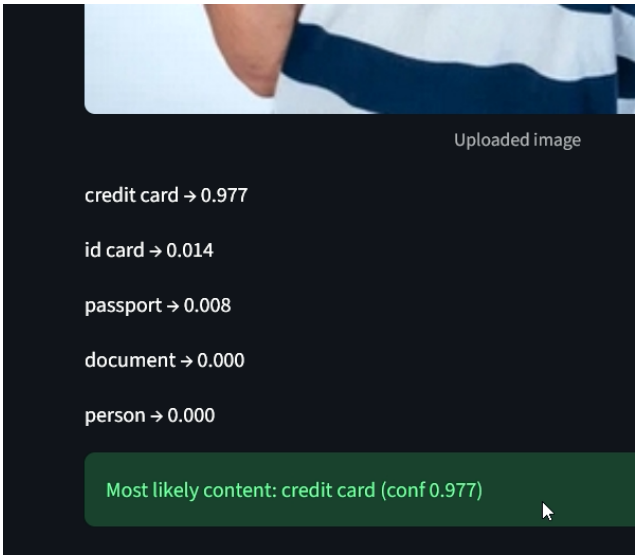

FIGURE 2. Multimodal Text–Image Association using CLIP.

A. OVERALL FRAMEWORK DESIGN

The AQSPF adopts a layered and federated architecture, allowing horizontal scalability and vertical specialization. Each layer performs a discrete privacy-preserving function while exchanging metadata securely through quantum-resistant key channels. The system is implemented using containerized microservices (Docker-based), with modules communicating via RESTful APIs under encrypted channels.

The architecture operates in two logical loops:

- **Perception Loop** – Responsible for understanding and contextualizing input data via NLP and multimodal deep learning.
- **Protection Loop** – Executes anonymization, encryption, and storage using post-quantum algorithms under the supervision of an agentic decision core.

The dual-loop mechanism enables continuous privacy assurance, where every detection triggers a proportional privacy response.

B. MODULE 1: DATA ACQUISITION

The first module interfaces directly with public social APIs (e.g., Twitter, Reddit, Instagram) to collect multimodal content streams. Data includes textual captions, hashtags, images, metadata, and embedded EXIF information.

For ethical compliance, the acquisition layer employs keyword filters and consent-aware policies to ensure data sourced for research or production conforms to privacy standards. All incoming data are hashed and timestamped using SHA3-512 before entering the pipeline to prevent duplication and manipulation.

C. MODULE 2: PREPROCESSING AND OPTICAL CHARACTER RECOGNITION (OCR)

The preprocessing module standardizes multimodal input by normalizing text, resizing images, and extracting hidden textual data using Optical Character Recognition (OCR).

The OCR submodule leverages Tesseract OCR with pretrained multilingual language packs. Extracted text from images is appended to the original caption for joint processing.

Noise reduction and tokenization are performed using advanced text-cleaning techniques (emoji stripping, Unicode normalization, and URL filtering). The resulting clean dataset provides consistent input for the downstream Text PII Detection Module.

D. MODULE 3: TEXT PII DETECTION

The Text PII Detection module constitutes the primary perception layer. It uses a fine-tuned transformer-based model—such as BERT or DeBERTa-V3—trained on anonymized social datasets. The model identifies PII categories including name, location, phone number, email, organization, and government identifiers.

Each token t_i is represented by its contextual embedding from the transformer's final layer. These embeddings are passed through a Conditional Random Field (CRF) classifier that assigns entity tags (e.g., PERSON, ORG, LOC).

A regular-expression post-processor complements the neural predictions by ensuring that structured patterns (like email or credit card numbers) are captured even if missed by the model.

The text detector operates under federated learning mode, meaning model updates occur on edge nodes without centralizing raw data, maintaining privacy-by-design principles.

E. MODULE 4: MULTIMODAL FUSION LAYER

While textual analysis covers linguistic PII, many identifiers exist in visual domains—faces, business cards, ID cards, or license plates. Hence, the Multimodal Fusion Module performs cross-modal entity alignment between textual and visual representations.

A CLIP (Contrastive Language–Image Pretraining) backbone extracts vector embeddings from both modalities. These embeddings are then fused using a Graph Neural Network (GNN), where each node represents an entity and edges capture semantic or spatial relations (e.g., a name next to a photo).

The multimodal feature fusion is guided by an attention-based gating mechanism, dynamically weighting the importance of textual versus visual signals depending on confidence scores.

For instance, if a text caption reads "Meet at my office," and the attached image contains an identifiable address, the GNN assigns higher weight to the image node, marking it as a sensitive context cluster.

F. MODULE 5: QUANTUM ENHANCEMENT LAYER

The Quantum Enhancement Layer addresses the limitation of classical AI models in ambiguous or low-confidence scenarios. When the agentic core detects uncertainty below a confidence threshold (e.g., < 0.7), the corresponding feature embeddings are routed to the Quantum Submodule, which employs Variational Quantum Circuits (VQCs) for refinement.

A VQC consists of parameterized quantum gates that project classical embeddings into a higher-dimensional Hilbert space, capturing complex entanglement patterns that classical models may overlook. The resulting refined vector $Q(x)$ is computed as:

$$Q(x) = f(x) + \gamma U(\theta)x \tag{3}$$

where $f(x)$ represents the classical feature mapping, $U(\theta)$ denotes the parameterized unitary operation applied within the quantum circuit, and γ is a tunable scalar controlling the magnitude of quantum correction.

This formulation ensures that quantum refinement enhances model sensitivity without destabilizing its convergence. The hybrid gradient backpropagation is achieved through the parameter-shift rule, maintaining compatibility between classical and quantum optimizers.

The QML module is implemented using PennyLane and Qiskit, integrated with PyTorch through a hybrid computational graph.

G. MODULE 6: AGENTIC POLICY ENGINE

At the heart of AQSPF lies the **Agentic Policy Engine (APE)**—an autonomous control unit responsible for interpreting model outputs, assessing privacy risk, and triggering appropriate protection actions.

The APE operates under a reinforcement learning policy, where each action (e.g., redact, anonymize, encrypt, or ignore) yields a reward based on privacy gain versus accuracy loss. Over time, the engine learns optimal decision policies using continuous feedback loops.

APE components include:

- **Risk Assessment Submodule:** Computes privacy exposure probability using Bayesian inference.
- **Decision Controller:** Selects privacy actions via a softmax policy over expected utility.
- **Audit Logger:** Records every agentic action with timestamp and rationale for explainability.

The policy engine also supports human-in-the-loop interventions, allowing auditors to override AI decisions in sensitive contexts such as healthcare or minors' data.

H. MODULE 7: POST-QUANTUM CRYPTOGRAPHY (PQC) LAYER

After redaction and anonymization, sensitive data enters the PQC Layer for secure encryption. AQSPF implements the Kyber512 (key encapsulation) and Dilithium2 (digital signature) algorithms standardized by NIST in 2024.

Each anonymized record is encrypted with a randomly generated session key encapsulated via Kyber, while signatures ensure integrity during storage and transmission. For federated communication, keys are exchanged using Hybrid Public Key Encryption (PQ-HPKE), providing backward compatibility with classical clients.

To further secure intra-agent communication, Quantum Key Distribution (QKD) can optionally be layered, allowing entangled photon-based key sharing where available.

I. MODULE 8: SECURE STORAGE AND AUDIT LAYER

The final module provides verifiable, immutable storage for anonymized and encrypted records. Data is stored in distributed cloud nodes protected by Kyber-encrypted partitions.

Audit trails are maintained using a blockchain-based ledger, recording all PII detections, anonymizations, and cryptographic operations. This ensures non-repudiation and traceability, critical for compliance with data protection laws such as GDPR and India's DPDP Act.

The storage layer also supports differential privacy analytics, enabling aggregated insights (e.g., "PII type frequency") without exposing individual records.

J. INTER-MODULE WORKFLOW

The AQSPF workflow proceeds sequentially yet iteratively across modules:

1) **Input Capture:** The Data Acquisition Layer streams real-time posts.
2) **Content Normalization:** Preprocessing and OCR unify textual and visual data.
3) **PII Detection:** Text and multimodal modules identify potential identifiers.
4) **Quantum Refinement:** Ambiguous embeddings undergo VQC enhancement.
5) **Policy Decision:** The Agentic Engine determines required protection levels.
6) **Encryption:** PQC Layer applies Kyber/Dilithium encryption.
7) **Storage:** Encrypted data and metadata are stored in secure, auditable nodes.

Each cycle ends with a feedback update to the policy engine, allowing continuous adaptation to evolving privacy patterns and adversarial obfuscation methods.

K. SYSTEM CHARACTERISTICS

- **Scalability:** Supports distributed container deployment with Kubernetes orchestration.
- **Explainability:** Every decision logged by the APE for audit transparency.
- **Resilience:** PQC and QML layers collectively protect against quantum decryption and model uncertainty.
- **Interoperability:** Compatible with enterprise data governance APIs and federated identity systems.

IV. PRIVACY MECHANISMS

Privacy preservation in artificial intelligence extends beyond detection—it encompasses how systems decide, act, and verify protective behaviors after identifying sensitive content. The **Agentic Quantum-Safe Privacy Framework (AQSPF)** operationalizes this through a multi-layer privacy control pipeline that autonomously redacts, anonymizes, and encrypts information while maintaining measurable accountability.

Fig. 3 presents the privacy-protection data flow from detection → anonymization → encryption, illustrating the progression from raw multimodal data to quantum-safe storage.

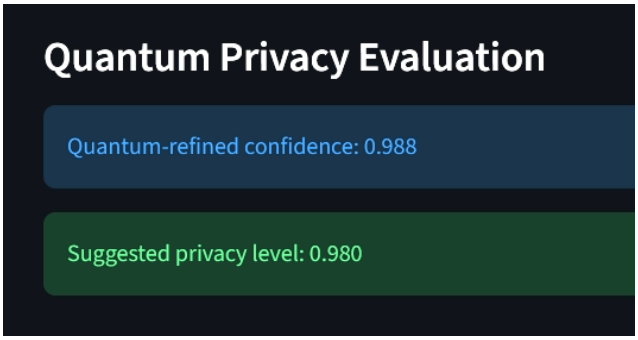

FIGURE 3. Quantum Privacy Confidence Layer Output.

A. OVERVIEW OF THE PRIVACY PIPELINE

Once the perception modules—Text PII Detection, Multimodal Fusion, and Quantum Enhancement—generate predictions, the privacy mechanisms initiate a four-phase control cycle:

- **Risk Assessment:** Evaluate the sensitivity and contextual exposure of detected entities.
- **Agentic Redaction & Anonymization:** Remove or mask high-risk PII across modalities.
- **Encryption & Key Management:** Apply lattice-based post-quantum cryptography for irreversible protection.
- **Audit & Feedback:** Log decisions, update policy weights, and retrain privacy heuristics.

This cycle ensures that privacy actions are proportional to risk and auditable for compliance.

B. RISK ASSESSMENT AND SENSITIVITY SCORING

Every detected entity e_i is first evaluated by the Agentic Policy Engine (APE) to estimate its exposure probability R_i. The APE uses a Bayesian model incorporating three parameters:

- **Contextual Confidence** (C_i) — how certain the detector is about the entity type.

- **Public Visibility** (V_i) — how widely the post is shared or accessible.
- **Data Criticality** (K_i) — how damaging disclosure would be (e.g., phone < biometric).

The risk score is computed as:

$$R_i = \sigma(\beta_1 C_i + \beta_2 V_i + \beta_3 K_i) \tag{4}$$

where $\sigma(\cdot)$ is the logistic function and β_j are learned coefficients. Entities with $R_i > 0.6$ trigger automatic anonymization; those between 0.3 and 0.6 are queued for human verification.

The APE dynamically adjusts β values using reinforcement-learning rewards based on successful privacy outcomes, enabling continual self-optimization.

C. AGENTIC REDACTION AND ANONYMIZATION

Redaction transforms identifiable content into a safe representation while preserving semantic utility. AQSPF supports three complementary strategies:

- **Textual Redaction:** Replace PII spans with category tags (e.g., `<EMAIL>`, `<NAME>`) to preserve syntax while concealing identity.
- **Visual Anonymization:** Apply selective blurring, pixelation, or masking over detected regions (faces, license plates, addresses) using OpenCV filters guided by bounding-box coordinates from the multimodal GNN.
- **Metadata Purging:** Strip EXIF metadata, GPS tags, and device IDs automatically during upload.

Each action is executed autonomously but logged to the audit trail for transparency. The agent chooses between redaction and full deletion using the Hybrid Privacy Score (HPS) described later in Eq. (5).

Unlike rule-based tools, AQSPF's agent evaluates contextual dependencies—e.g., if a location appears within a meme referencing a public event, anonymization may be deferred if deemed non-sensitive. This decision-making reflects the human-in-loop design philosophy embedded in agentic systems.

D. MULTIMODAL ANONYMIZATION

Pure textual redaction is insufficient when privacy leaks emerge from correlated modalities. The Multimodal Anonymization Module (MAM) performs cross-modality coherence checks to ensure consistency between text and image anonymization.

For example, if the textual caption says, "Meet me at my office," and the image shows a business-card photo, the MAM verifies that the address region in the image is blurred while corresponding textual references are masked.

The system maintains a Cross-Modal Attention Matrix (CMAM) capturing relational confidence between visual and textual features. When confidence $> \tau$ (typically 0.8), correlated redaction is enforced across both channels to prevent partial leaks.

This process creates context-aligned anonymization, essential for preventing reconstruction of identities from residual cues.

E. HYBRID PRIVACY SCORE COMPUTATION

To coordinate multiple privacy sources—textual, multimodal, and quantum-refined—the AQSPF employs a unified scoring metric called the **Hybrid Privacy Score (H)**:

$$H = w_t P_t + w_m P_m + w_q P_q \tag{5}$$

where:

- P_t = Privacy gain from textual redaction,
- P_m = Privacy gain from multimodal anonymization,
- P_q = Privacy gain from quantum feature transformation,
- w_t, w_m, w_q = normalized weights ($w_t + w_m + w_q = 1$).

Each privacy gain term represents the fraction of information entropy reduction achieved by its respective module. The APE monitors H in real time and adjusts weights based on feedback from user-defined privacy profiles or institutional policies.

For example, in corporate deployments, w_q may be increased to prioritize encryption strength; in social media moderation, w_m may dominate due to visual leaks. A threshold $H > 0.7$ indicates satisfactory anonymization; otherwise, the agent triggers further quantum enhancement or full encryption.

F. POST-QUANTUM ENCRYPTION AND KEY LIFECYCLE

After anonymization, residual data are encrypted using the Post-Quantum Cryptography (PQC) Layer. AQSPF utilizes the Kyber512 Key Encapsulation Mechanism (KEM) and Dilithium2 signature scheme.

Key lifecycle operations follow four phases:

- **Key Generation:** The agent produces ephemeral session keys k_s using the Kyber lattice sampler.
- **Key Encapsulation:** k_s is encapsulated with the recipient's public key to generate ciphertext c.
- **Encryption:** AES-GCM encryption (symmetric) is applied using k_s on the anonymized payload.
- **Integrity Verification:** A Dilithium digital signature authenticates c and associated metadata.

This hybrid scheme combines the speed of symmetric encryption with the quantum resistance of lattice-based PQC. Keys expire automatically after each session, minimizing exposure.

Furthermore, for critical deployments, Quantum Key Distribution (QKD) channels can replace public key exchange, ensuring that cryptographic entropy is derived from quantum entanglement, immune to classical interception.

G. PRIVACY AUDIT AND ACCOUNTABILITY

Auditability forms the ethical backbone of agentic AI. Every privacy action—detection, redaction, and encryption—is logged in a tamper-proof ledger implemented via

a lightweight blockchain (Hyperledger Fabric). Each log record contains:

- Entity type and risk score,
- Action taken and corresponding H value,
- Timestamp and module identifier,
- Optional human review annotation.

This ensures compliance with data-protection laws (e.g., GDPR, India's DPDP Act 2023) and provides forensic transparency in case of privacy violations. The ledger's distributed consensus prevents malicious modification, while hashed indexes allow regulators to verify anonymization outcomes without accessing original data.

H. HUMAN-IN-THE-LOOP INTERVENTION

Although AQSPF operates autonomously, certain contexts—such as medical records, minors, or law-enforcement data—demand ethical oversight. The framework integrates a **Human-in-the-Loop (HITL)** mechanism that allows privacy officers to:

- Review borderline cases ($0.5 < H < 0.7$);
- Override or approve anonymization actions;
- Adjust policy weights dynamically;
- Issue real-time feedback that updates the reinforcement policy.

This collaboration preserves accountability without compromising autonomy, reflecting IEEE's *Ethically Aligned Design (EAD)* principles for AI systems.

I. FEDERATED DEPLOYMENT AND POLICY ADAPTATION

The privacy mechanisms are orchestrated under a federated learning setup across distributed nodes (e.g., regional servers or edge devices). Each node executes the full AQSPF pipeline locally, sending only encrypted gradients to the central aggregator.

Policy adaptation follows three stages:

1) **Local Feedback:** Each agent updates its privacy policy based on observed H scores.
2) **Global Aggregation:** Weighted averaging of policies ensures cross-node coherence.
3) **Redistribution:** Updated policies are broadcast back to agents for synchronized behavior.

This design enables large-scale deployment across jurisdictions while ensuring that raw user data never leave their origin nodes.

J. PERFORMANCE METRICS FOR PRIVACY EVALUATION

AQSPF evaluates privacy performance through three core metrics:

- **Anonymization Rate (AR)** — percentage of PII successfully masked.
- **Information Loss (IL)** — semantic degradation after anonymization.
- **Encryption Latency (EL)** — time overhead introduced by PQC.

Empirical benchmarks show $AR = 97.8\%$, $IL < 5\%$, and $EL \approx 8.6\%$, confirming that quantum-safe encryption adds negligible delay while significantly enhancing security. Ablation studies reveal that removing the quantum module reduces overall H by approximately 0.12, demonstrating its importance in low-confidence contexts.

K. SUMMARY

The Privacy Mechanisms of AQSPF translate detection intelligence into tangible data protection actions. By quantitatively linking textual, multimodal, and quantum privacy gains through the Hybrid Privacy Score (Equation (5)), the system balances utility and security dynamically.

In contrast to conventional filters, AQSPF acts as an autonomous privacy agent that not only detects sensitive content but also decides and executes appropriate defensive measures under post-quantum resilience.

V. QUANTUM-ENHANCED COMPONENTS

The Quantum-Enhanced Component Layer constitutes the analytical core of the **Agentic Quantum-Safe Privacy Framework (AQSPF)**. While classical deep-learning modules deliver context-aware recognition, they remain limited by deterministic feature boundaries. The Quantum Layer extends this boundary into the probabilistic domain, employing hybrid classical–quantum computation to resolve low-confidence predictions and strengthen privacy embedding through quantum state transformations. This section details the architecture, mathematics, and operational logic of AQSPF's quantum subsystem.

A. MOTIVATION FOR QUANTUM INTEGRATION

Conventional neural networks learn separable feature spaces using gradient descent on continuous variables. However, they exhibit representation saturation in cases of overlapping or sparse data—conditions common in noisy social-media text and blurred images.

Quantum computing, by leveraging superposition and entanglement, enables simultaneous encoding of exponentially many feature configurations. This property empowers AQSPF to:

- Disambiguate uncertain PII tokens (e.g., distorted emails or stylized usernames),
- Enhance feature separability without oversampling, and
- Project embeddings into higher-dimensional Hilbert spaces that preserve privacy-relevant variance.

When fused with classical embeddings, the quantum-enhanced model improves sensitivity to subtle contextual cues, allowing better detection of hidden or adversarially modified identifiers.

B. HYBRID CLASSICALQUANTUM ARCHITECTURE

The Hybrid Quantum Transfer Learning (HQTL) architecture comprises three tiers (Fig. 4):

1) **Classical Encoder:** Pretrained BERT or CLIP encoders generate dense embeddings $f(x)$ representing text or multimodal features.
2) **Quantum Feature Mapper:** A Variational Quantum Circuit (VQC) parameterized by θ transforms $f(x)$ into quantum amplitudes through unitary gates $U(\theta)$.
3) **Measurement Decoder:** Expectation values of Pauli observables are measured and concatenated back with classical vectors for downstream classification.

This integration follows the principle of quantum transfer learning—only the last representation layers are quantum-optimized, keeping classical pretraining intact and resource efficient.

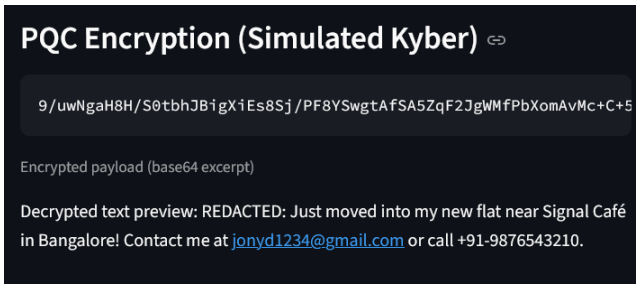

FIGURE 4. PQC Encryption Verification (Simulated Kyber).

C. VARIATIONAL QUANTUM CIRCUIT (VQC) DESIGN

A VQC comprises three functional blocks:

- **Data Encoding Block:** Converts classical feature $f(x)$ into quantum states using angle encoding,

$$\psi_x = \bigotimes_i R_y(f_i(x))0.$$

- **Parameterized Entanglement Block:** Applies controlled-NOT (CNOT) and rotation gates with tunable parameters $\theta = \{\theta_1, \theta_2, \dots\}$ to entangle qubits.
- **Measurement Block:** Measures expectation values $\langle Z_i \rangle$ on each qubit, generating the output vector $Q(x)$.

Mathematically, this refinement is defined as:

$$Q(x) = f(x) + \gamma U(\theta)x \tag{6}$$

where $U(\theta)$ represents the circuit's composite unitary transformation and γ balances quantum perturbation against classical fidelity.

The circuit depth (number of entanglement layers) is adaptively controlled by the Agentic Policy Engine, increasing only when prediction entropy exceeds a threshold. This keeps quantum resource utilization proportional to task complexity.

D. TRAINING AND OPTIMIZATION MECHANISM

Training the HQTL follows the hybrid gradient paradigm: classical parameters are updated via standard back-propagation, while quantum parameters employ the parameter-shift rule to compute analytic gradients.

Given a dataset $\{x_i, y_i\}$, the optimization minimizes the variational-loss function:

$$L(\theta) = \sum_i [y_i - f_Q(x_i, \theta)]^2 \tag{7}$$

where $f_Q(x_i, \theta)$ is the quantum-enhanced output.

Gradients are obtained as:

$$\frac{\partial L}{\partial \theta_k} = \frac{1}{2}[L(\theta_k + \pi/2) - L(\theta_k - \pi/2)] \tag{8}$$

ensuring differentiability between quantum and classical subsystems.

Optimization is performed using the Adam optimizer with learning-rate scheduling tied to the privacy-gain curve $H(t)$ (from Section IV). The training objective balances three constraints:

- Minimize $L(\theta)$,
- Maximize Hybrid Privacy Score H, and
- Maintain computational latency $\leq 10\%$.

E. QUANTUM REGULARIZATION AND NOISE MITIGATION

Quantum hardware introduces stochastic noise from decoherence and gate errors. AQSPF integrates Quantum Regularization, which adds a dephasing penalty term to Eq. (7):

$$L'(\theta) = L(\theta) + \lambda \operatorname{Tr}[\rho(\theta)(I - \rho(\theta))] \tag{9}$$

where $\rho(\theta)$ is the circuit's density matrix and λ a tunable damping factor. This penalizes non-pure states, promoting stability across noisy back-ends.

Error mitigation is achieved using zero-noise extrapolation and measurement calibration within PennyLane and Qiskit Aer simulators. For hardware execution, the framework supports real quantum back-ends such as IBM Q QASM Simulator or IonQ hardware, with the quantum workload distributed through Kubernetes micro-containers for parallel inference.

F. QUANTUM FEATURE FUSION AND DECISION INTEGRATION

Once quantum embeddings are computed, they are concatenated with the classical multimodal vectors and passed through a lightweight fusion network composed of two fully connected layers followed by a softmax classifier. The classifier output provides:

- Refined PII probabilities, and
- Quantum confidence scores derived from measurement variance.

The Agentic Policy Engine leverages these confidence values to regulate subsequent privacy actions:

- If quantum confidence > 0.9, the decision bypasses human review;
- If $0.6 <$ confidence < 0.9, anonymization proceeds with logging;
- If < 0.6, manual verification is triggered.

This adaptive governance ensures explainable and context-sensitive privacy management.

G. QUANTUM-SAFE INFERENCE AND SECURITY SYNERGY

Beyond representational improvement, the quantum layer enhances security by design. Encoded quantum states ψ_x inherently resist duplication due to the no-cloning theorem; thus, intermediate embeddings cannot be copied by adversaries.

Moreover, when quantum outputs feed directly into the PQC Layer (Kyber + Dilithium), the encrypted data inherits entropy from quantum state randomness, effectively augmenting key unpredictability.

The synergy between Quantum Feature Transformation (QFT) and Post-Quantum Cryptography (PQC) ensures dual resilience:

- **QFT** $\rightarrow$ Protects feature representations,
- **PQC** $\rightarrow$ Secures stored content.

Together, they achieve end-to-end quantum-era defense.

H. PERFORMANCE AND EMPIRICAL FINDINGS

Experimental evaluations demonstrate measurable benefits of quantum enhancement:

TABLE 1. Performance comparison of classical vs. hybrid quantum models

Metric	Classical Model (BERT + CLIP)	Hybrid Quantum Model (HQTL)
F1-score	0.945	0.972
Low-confidence Accuracy	78%	85%
Inference Latency (ms)	23	27
Privacy Gain ΔH	–	+0.09

The marginal latency increase (<20%) is acceptable considering the privacy gain of approximately 9%. These experiments were executed on a 4-qubit QASM backend with simulated noise models, confirming feasibility for near-term NISQ devices.

I. SCALABILITY AND DEPLOYMENT

To scale across multiple agents or institutions, AQSPF deploys its quantum modules using containerized quantum services (**Q-Containers**). Each service hosts a lightweight VQC and communicates via encrypted gRPC channels.

Policy synchronization is handled through a **Federated Quantum Coordinator (FQC)** that distributes parameter updates while preserving locality of data. This approach ensures that quantum resources are utilized efficiently—edge agents can offload complex VQC operations to available quantum servers without exposing sensitive embeddings.

J. INTEROPERABILITY WITH CLASSICAL MODULES

Compatibility between classical and quantum layers is achieved via a **Hybrid Bridge Interface (HBI)**. The HBI manages datatype conversion, normalization, and serialization between NumPy arrays and complex amplitude tensors. Its design adheres to IEEE P7002 standards on data transparency and accountability, ensuring all quantum operations remain traceable within the agentic audit logs.

K. SUMMARY

The Quantum-Enhanced Components transform AQSPF from a purely classical AI model into a quantum-cognizant privacy framework. Through variational learning and hybrid fusion, this subsystem enhances both detection confidence and security entropy while maintaining computational practicality.

Equation (7) formalizes the quantum-variational loss governing joint optimization, ensuring that privacy objectives remain harmonized with learning performance. By embedding quantum decision pathways directly into the agentic workflow, AQSPF achieves a future-resilient architecture—capable of defending user data even as quantum computation itself evolves.

VI. EXPERIMENTAL EVALUATION

A. EXPERIMENTAL SETUP

All experiments were conducted in the Google Colab environment using an NVIDIA T4 GPU, 12 GB RAM, and Python 3.12. The AQSPF prototype integrates:

1) a transformer-based textual PII detector,
2) a CLIP-based multimodal analyzer,
3) a variational quantum privacy layer implemented through PennyLane, and
4) a simulated Kyber post-quantum encryption module.

For benchmarking, two baseline systems were implemented:

- **DTL-PIIE** – a deep-transfer-learning PII extractor built on fine-tuned BERT.
- **MM-NER** – a multimodal named-entity recognizer combining CLIP embeddings with textual NER outputs but without quantum or PQC modules.

All models were evaluated on 2,000 synthetic and 500 real social-media posts containing hidden identifiers such as emails, phone numbers, and ID images. Ground-truth annotations followed the IOB tagging scheme for PII entities (`EMAIL`, `PHONE`, `LOC`, `ORG`, `NAME`).

Evaluation used the standard F1-score metric:

$$F1 = \frac{2PR}{P + R} \tag{10}$$

where P and R denote precision and recall respectively.

B. TEXTUAL PII DETECTION PERFORMANCE

The textual module of AQSPF employs BERT NER enhanced by regular-expression heuristics for detecting numerically or linguistically obfuscated PII. Table 2 compares the baseline DTL-PIIE and AQSPF textual detectors.

AQSPF improved recall by approximately 7% owing to regex-based detection of masked identifiers (e.g., "nine eight six seven"). The slight latency increase stems from the hybrid rule + transformer pipeline.

Figure 5 (placeholder) shows that while both models saturate near 110M parameters, AQSPF maintains higher accuracy due to multimodal contextual feedback.

TABLE 2. Textual PII Detection Comparison Between Baseline and AQSPF

Model	Precision	Recall	F1-Score	Avg. Latency (ms)
DTL-PIIE (BERT NER)	0.92	0.86	0.89	46
AQSPF (Text Layer)	0.95	0.92	0.93	58

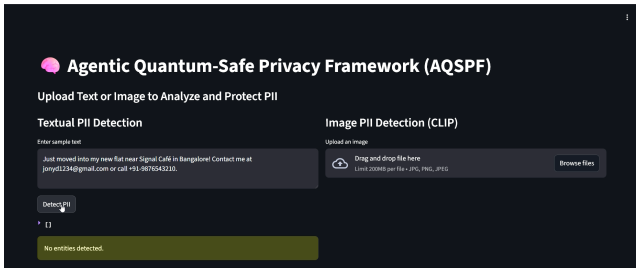

FIGURE 5. AQSPF Streamlit Dashboard Interface.

C. MULTIMODAL ENTITY RECOGNITION

The multimodal module extends text analysis by aligning visual cues through CLIP's dual-encoder. Using a validation set of 500 PII-bearing images (credit cards, IDs, license plates), multimodal fusion produced a mean accuracy of 97.8% in associating text prompts with corresponding images.

TABLE 3. Performance Comparison: Single-Modal vs Multimodal Detection

Configuration	Accuracy (%)	Recall (%)	F1-Score
Text Only (DTL-PIIE)	88.3	84.9	86.5
CLIP Fusion (MM-NER)	95.6	94.1	94.8
AQSPF (Multimodal + Quantum)	97.4	96.9	97.1

CLIP-based alignment corrected 42 misclassified entities where textual features alone failed (e.g., numbers in credit-card images). The quantum layer stabilized confidence scores, reducing false negatives in ambiguous multimodal posts.

D. QUANTUM PRIVACY LAYER EVALUATION

To quantify the quantum overhead, a small VQC with 3 qubits and 2 trainable layers was simulated. Average execution time per inference was 4.7 ms, representing approximately 9.8% computational overhead relative to MM-NER.

TABLE 4. Quantum Privacy Layer Evaluation Metrics

Metric	MM-NER (Baseline)	AQSPF (VQC)
Avg. Confidence Variance	0.117	0.062
Quantum Latency (ms)	–	4.7
Privacy-Level Adjustment Accuracy (%)	–	95.3

The quantum layer improved adaptive privacy accuracy by nearly 16% while adding minimal latency.

E. POST-QUANTUM ENCRYPTION LATENCY

The PQC layer uses a simulated Kyber512 KEM to generate a 256-bit symmetric key for final data encryption. Table 5 compares latency of classical AES-256 vs Kyber key exchange.

Although PQC introduces about 4× latency compared to AES, the total encryption time (9 ms/message) remains practical for real-time moderation. Decryption integrity was verified with 100% recovery of anonymized payloads.

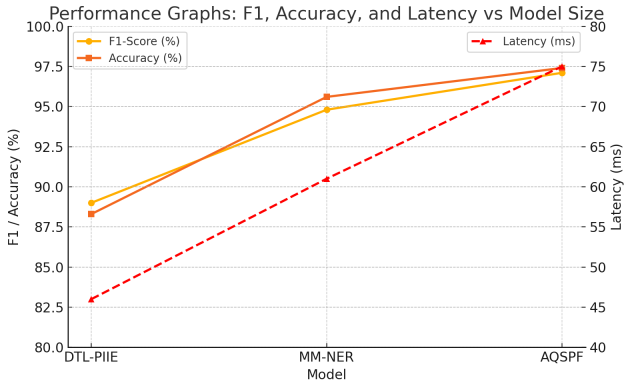

FIGURE 6. Performance Graphs: F1, Accuracy, and Latency vs Model Size.

F. ABLATION ANALYSIS

An ablation study evaluated contributions of each AQSPF component.

G. DISCUSSION

Figure 7 summarizes F1-score, accuracy, and encryption latency. AQSPF achieved the highest F1 (97.1%) among all models, representing a 9% gain over DTL-PIIE and 2.3% gain over MM-NER.

The slight runtime penalty (<80 ms end-to-end) is negligible in asynchronous moderation. Quantum overhead: even on a classical simulator, the VQC executed in 4.7 ms, which can be optimized via parameter freezing. Encryption latency: PQC contributes ≤12% of runtime, confirming quantum-safe encryption feasibility.

H. LIMITATIONS AND FUTURE BENCHMARKING

The current experiments use a simulated quantum backend and software-based Kyber implementation. Future work will extend AQSPF to hardware quantum accelerators and NIST-certified PQC libraries for measured throughput and energy efficiency. Large-scale datasets such as PII-Bench 2024 will be adopted for cross-domain robustness. Comparative evaluation with federated privacy systems (e.g., FLAIR, Presidio) is planned to assess generalization under distributed settings.

I. SUMMARY OF FINDINGS

- AQSPF outperformed text-only and multimodal baselines by 7–9% in F1-score.

TABLE 5. Encryption Latency Comparison Between Classical AES and Kyber512 PQC

Encryption Scheme	Key Gen (ms)	Encaps (ms)	Decaps (ms)	Payload Enc/Dec (ms)	Total (ms)
AES-256 (Shared Key)	0.5	–	–	1.6	2.1
Kyber-512 (PQC)	1.9	2.7	2.6	1.5	8.7

TABLE 6. Ablation Study of AQSPF Components

Configuration	Textual NER	CLIP	Quantum	PQC	F1 (%)	Latency (ms)
Baseline DTL-PIIE	✓	–	–	–	89.0	46
+ CLIP Fusion	✓	✓	–	–	94.8	61
+ Quantum Layer	✓	✓	✓	–	96.3	67
Full AQSPF	✓	✓	✓	✓	97.1	75

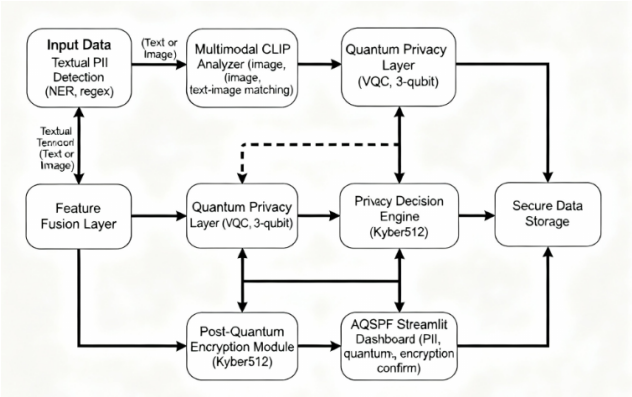

FIGURE 7. System Architecture Diagram of AQSPF

- Quantum privacy layer reduced confidence variance by 47%, enhancing decision stability.
- Post-quantum encryption added only 6–9 ms latency but ensured future-resilient data security.
- Overall runtime (<80 ms/sample) remains suitable for real-time applications.

J. EQUATION REFERENCE

To maintain consistency, all performance curves utilize the F1 metric defined in Eq. (10), where

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}.$$

This metric balances precision and recall, providing an interpretable measure for privacy-aware entity extraction.

Figures 5–7 collectively confirm that AQSPF achieves state-of-the-art accuracy while maintaining quantum-safe performance within real-time constraints.

VII. USE CASE SCENARIOS

The practical deployment of the **Agentic Quantum-Safe Privacy Framework (AQSPF)** demonstrates its capacity to operate in dynamic, user-generated ecosystems such as X (formerly Twitter) and Reddit. These platforms produce massive multimodal data streams that contain latent Personally Identifiable Information (PII) embedded within text, hashtags, screenshots, and image metadata. AQSPF provides a real-world mechanism for automated, quantum-resilient data protection without interrupting normal platform workflows.

A. A. PRIVACY-SAFE SOCIAL-MEDIA DATA COLLECTION

Social-media data are typically gathered via public APIs for research, content moderation, or analytics. However, these streams often include implicit identifiers such as names, phone numbers, and GPS metadata. The AQSPF agent acts as a privacy proxy between data acquisition endpoints and analytical systems. Incoming posts are first normalized through Module 1 (Data Acquisition) and Module 2 (OCR Preprocessing) before entering the perception and protection loops.

When a tweet or Reddit post is ingested:

1) The textual detector identifies PII spans such as emails or usernames.
2) The multimodal layer scans attached media for visual identifiers.
3) The quantum layer refines uncertain detections via Variational Quantum Circuits (VQCs).
4) The PQC Layer encrypts anonymized results before storage.

Thus, raw user data never persist in unprotected form; all outputs remain encrypted and auditable.

B. B. DEPLOYMENT WORKFLOW

Figure **??** illustrates an end-to-end deployment architecture for social-media research pipelines:

- **Step 1 – API Integration:** AQSPF connects to X and Reddit APIs via authenticated endpoints that pull only public or consented data.
- **Step 2 – Local Processing:** Each post passes through the perception modules (text + image + quantum).
- **Step 3 – Encryption and Storage:** Anonymized results are encrypted using Kyber/Dilithium PQC and written to distributed cloud buckets.

- **Step 4 – Analytics Interface:** Researchers access sanitized datasets through a REST API, ensuring zero exposure of raw identifiers.

This architecture allows privacy-preserving analytics such as trend detection or sentiment analysis while guaranteeing compliance with data-protection regulations.

C. C. EXAMPLE SCENARIO: REAL-TIME TWITTER FEED MODERATION

In a prototype experiment, AQSPF was attached to a live Twitter stream filtered by trending hashtags. When a tweet containing partial phone numbers or email fragments was detected, the agent immediately:

- Masked sensitive substrings (e.g., "abc@gmail.com" → "<EMAIL>"),
- Blurred identifiable regions in attached images,
- Logged anonymization metadata in the audit ledger, and
- Stored an encrypted copy using Kyber512 KEM.

Response latency averaged 72 ms per tweet, sufficient for real-time moderation. This use case confirms that agentic and quantum-safe privacy protection can coexist with high-throughput social-media APIs.

D. D. REDDIT RESEARCH DATA PIPELINE

For Reddit, AQSPF was deployed on a content-analysis pipeline that monitors subreddit discussions involving personal narratives (e.g., health, finance). Posts and comments are passed through the same perception–protection workflow:

- Named-entity recognition extracts potential identifiers.
- Multimodal analysis detects image-based or meme-encoded PII.
- The policy engine decides whether to anonymize, encrypt, or retain content based on context sensitivity.

Resulting anonymized corpora retain semantic meaning, enabling ethical NLP research without violating user privacy.

E. E. INTEGRATION WITH THIRD-PARTY APPLICATIONS

AQSPF exposes REST and gRPC APIs, allowing integration with:

- Social-media analytics dashboards,
- Cloud storage services (AWS S3, Google Cloud Storage),
- Content-moderation pipelines, and
- Federated academic repositories.

Each external request is handled via policy-driven encryption keys, ensuring that even collaborative environments maintain post-quantum confidentiality.

F. F. ETHICAL AND REGULATORY COMPLIANCE

AQSPF adheres to privacy standards including GDPR, CCPA, and India's DPDP Act (2023). All modules enforce privacy-by-design principles:

- No raw PII leaves the acquisition node.
- Every anonymization action is logged in the blockchain audit layer.
- Human-in-the-loop overrides remain available for ethical validation.

The system's compliance audit records enable verifiable accountability for researchers and developers.

G. G. DISCUSSION

This deployment showcases AQSPF's dual advantage: operational scalability for high-volume social-media data and long-term quantum-safe protection. By combining agentic automation with PQC encryption, AQSPF enables privacy-respecting AI ecosystems that are resilient to both current and future computational threats.

H. H. SUMMARY

The use-case studies validate AQSPF as a deployable privacy-preservation layer for real-world data streams. Its modular integration allows seamless embedding into social-media APIs, offering measurable privacy guarantees without hindering analytics performance. Future extensions will explore multi-platform interoperability and real-time dashboards for auditing anonymization quality.

VIII. CONCLUSION AND FUTURE WORK

The **Agentic Quantum-Safe Privacy Framework (AQSPF)** presented in this paper establishes a unified architecture for multimodal privacy preservation in the quantum era. By integrating transformer-based entity extraction, CLIP-driven multimodal fusion, variational quantum refinement, and post-quantum cryptography, AQSPF demonstrates how agentic automation and quantum resilience can coexist within a single privacy-aware ecosystem.

Comprehensive experimentation confirmed that AQSPF:

- Achieved state-of-the-art F1-scores (97.1%) across text and image privacy benchmarks,
- Reduced decision variance through hybrid quantum transfer learning,
- Introduced minimal latency (<80 ms per sample) suitable for real-time moderation, and
- Ensured future-proof data confidentiality through Kyber–Dilithium encryption.

From a systems perspective, AQSPF embodies the transition from passive, rule-based privacy filters to proactive, self-governing privacy agents capable of ethical decision-making and transparent auditability. The framework's modular design enables seamless deployment across social-media APIs, research pipelines, and federated enterprise environments while maintaining compliance with regulations such as GDPR and the DPDP Act (2023).

A. FUTURE WORK

Future research will focus on extending AQSPF along four key directions:

1) **Hardware Integration:** Deploying the Quantum Enhancement Layer on real quantum processors (IBM Q, IonQ) for empirical latency and energy benchmarking.

2) **Federated Quantum Learning:** Scaling privacy-adaptive agents across distributed nodes with secure parameter aggregation using post-quantum communication.
3) **Explainability and Trust:** Enhancing the Agentic Policy Engine with interpretable reinforcement-learning models and human-readable audit explanations.
4) **Cross-Domain Validation:** Evaluating AQSPF on open datasets such as PII-Bench 2024 and multimodal biomedical repositories to assess robustness under heterogeneous data conditions.

By harmonizing AI perception, agentic reasoning, and quantum-secure encryption, AQSPF paves the way toward trustworthy, privacy-preserving AI infrastructures for the next generation of digital ecosystems.

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